

# Infosys Springboard Virtual Internship 7.0 Completion Report

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**Batch Number:** 1

**Start date:** 23-06-2026

**Names:**

- 1 Kancharla Poojitha
- 2 Vempati Joshitha
- 3 Harshini A P
- 4 Himanshu
- 5 Nithin Kumar

Internship Duration: 40-day (12 Weeks)

## 1. Project Title

### AI-BASED INDUSTRIAL MAINTENANCE AND PROCESS OPTIMIZATION

## 2. Project Objective

AI Based Industrial Maintenance and Process Optimization is an AI-driven system designed to predict potential machine failures and provide intelligent insights for proactive maintenance. Its core objectives are to:

- Predict potential machine failures using machine learning models trained on industrial machine data.
- Analyze machine operating conditions using parameters such as air temperature, process temperature, rotational speed, torque, tool wear, and machine type.
- Estimate machine failure probability and classify the machine into different risk levels.
- Provide intelligent maintenance recommendations based on predicted risk and observed machine conditions.
- Monitor prediction history and machine-risk trends through process intelligence and dashboard analytics.
- Support proactive maintenance decisions by transforming machine sensor data into meaningful and actionable insights.

The project aims to move from reactive maintenance to predictive and data-driven maintenance, helping identify potential machine failures earlier, reduce unexpected downtime, and improve maintenance planning.

## 3. Project Description in Detail

AI Based Industrial Maintenance and Process Optimization system is an AI-powered web application designed to predict machine failures and provide intelligent maintenance insights using industrial machine operating data.

The system uses the AI4I 2020 Predictive Maintenance Dataset and processes machine operating parameters, after which it:

1. Performs data preprocessing and validation by checking missing values, duplicate records, data types, and feature distributions.
2. Performs exploratory data analysis (EDA) to understand feature distributions, correlations, machine failure patterns, and relationships between operating parameters.
3. Prepares the dataset for machine learning by selecting relevant features and applying One-Hot Encoding to the categorical machine Type feature.
4. Trains and evaluates different machine learning and deep learning models, including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, and MLP.
5. Uses the selected production model to predict:
  - Machine failure status
  - Failure probability
6. Converts the predicted failure probability into meaningful risk levels:
  - Low
  - Moderate

- High
- Critical

7. Performs additional sensor-condition analysis to identify abnormal operating conditions such as high torque, excessive tool wear, abnormal rotational speed, and temperature-related conditions.

8. Generates maintenance recommendations based on the predicted failure risk and machine operating conditions.

9. Stores prediction results and provides process intelligence through dashboard analytics, including:

- Prediction history
- Risk distribution
- Machine-type distribution
- Average failure probability
- Failure-probability trends
- Recent and high-risk predictions
- Model performance information

10. Provides a web-based application with user authentication, machine failure prediction, database storage, REST APIs, and dashboard-based process intelligence.

The project combines AI-based predictive maintenance with process intelligence to transform raw machine operating data into failure predictions, risk assessments, maintenance recommendations, and analytical insights, supporting more proactive and informed industrial maintenance decisions.

#### 4. Timeline Overview

| Week          | Activities Planned                                                                              | Activities Completed                                                                                                    |
|---------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Week 1</b> | Requirement study and understanding the Predictive Maintenance and Process Intelligence problem | Project requirements studied, problem definition and objectives finalized                                               |
| <b>Week 2</b> | Dataset evaluation and technology stack planning                                                | AI4I 2020 Predictive Maintenance Dataset selected and project environment configured                                    |
| <b>Week 3</b> | Dataset loading and initial data inspection                                                     | Dataset loaded, shape, data types, columns, and basic statistics analysed                                               |
| <b>Week 4</b> | Data quality checking and preprocessing                                                         | Missing values, duplicate records, categorical features, and data consistency checked                                   |
| <b>Week 5</b> | Exploratory Data Analysis                                                                       | Feature distributions, skewness, correlations, failure patterns, and machine-type analysis completed                    |
| <b>Week 6</b> | Feature preparation and model development                                                       | Relevant features selected, categorical Type feature encoded using One-Hot Encoding, and dataset prepared for modelling |
| <b>Week 7</b> | Machine learning model training and evaluation                                                  | Logistic Regression, Decision Tree, Random Forest, Gradient Boosting and other models trained and evaluated             |

|                |                                                           |                                                                                                                       |
|----------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Week 8</b>  | Model comparison and production model selection           | Models compared using Accuracy, Precision, Recall, F1-score, ROC-AUC and confusion matrix; production model selected  |
| <b>Week 9</b>  | Backend and prediction API development                    | Flask REST API integrated with the trained model for machine failure prediction                                       |
| <b>Week 10</b> | Risk assessment, recommendations and database integration | Risk levels, sensor observations, maintenance recommendations, user authentication and prediction history implemented |
| <b>Week 11</b> | Dashboard and process intelligence implementation         | Dashboard analytics, risk distribution, prediction trends, machine-type analysis and model information integrated     |
| <b>Week 12</b> | Testing, documentation and final integration              | End-to-end testing completed, project documentation prepared, and complete system made ready for evaluation           |

### 5a. Key Milestones

| Milestone                            | Description                                                                                                                                                 | Date Achieved      |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Project Kick off</b>              | Defined project objectives, scope, responsibilities, and success criteria.                                                                                  | <b>25 Jun 2025</b> |
| <b>Dataset Selection &amp; Setup</b> | Evaluated available datasets and selected the <b>AI4I 2020 Predictive Maintenance Dataset</b> . Project environment and required libraries were configured. | <b>09 Jul 2025</b> |
| <b>ML Model Development</b>          | Developed and evaluated multiple machine learning models for predicting machine failure using the prepared dataset                                          | <b>16 Jul 2025</b> |
| <b>System Prototype</b>              | Integrated the trained machine learning model with the Flask backend and implemented the machine failure prediction API.                                    | <b>23 Jul 2025</b> |
| <b>Final Submission</b>              | Completed the integrated Predictive Maintenance and Process Intelligence system along with project documentation and testing.                               | <b>06 Aug 2026</b> |
| <b>Final Presentation</b>            | Presented the project covering the problem, methodology, machine learning models, system architecture, results, and application functionality.              | <b>13 Aug 2026</b> |

## 5b. Project execution details

The AI Based Industrial Maintenance and Process Optimization project was developed through a structured approach.

The project began with **requirement analysis and dataset selection**, where the AI4I 2020 Predictive Maintenance Dataset was selected.

Next, **data preprocessing and EDA** were performed, including missing-value and duplicate checks, feature analysis, correlation analysis, and failure-pattern analysis. Categorical machine Type was handled using **One-Hot Encoding**.

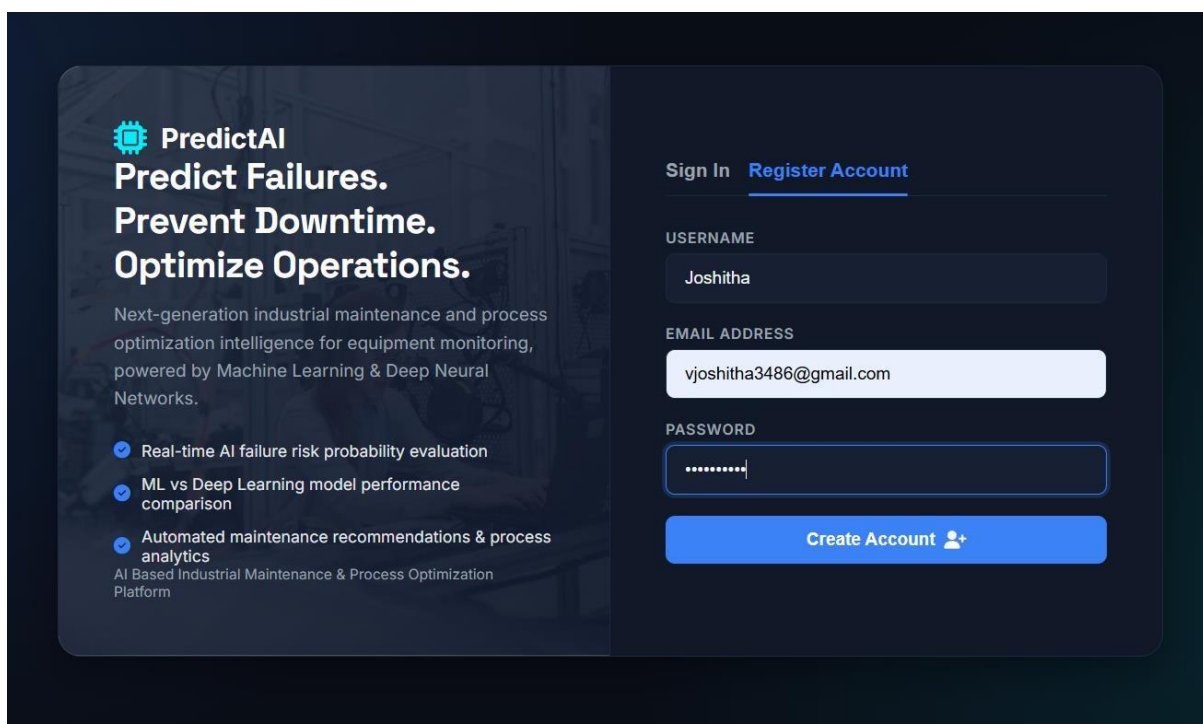
Multiple **machine learning models** were then trained and evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix. A suitable production model was selected based on the evaluation results.

The selected model was integrated into a **Flask REST API**, where machine parameters are used to predict failure probability and assign **Low, Moderate, High, or Critical** risk levels.

Finally, **maintenance recommendations, prediction history, authentication, database integration, and dashboard analytics** were implemented. The complete system was tested, refined, documented, and prepared for final evaluation.

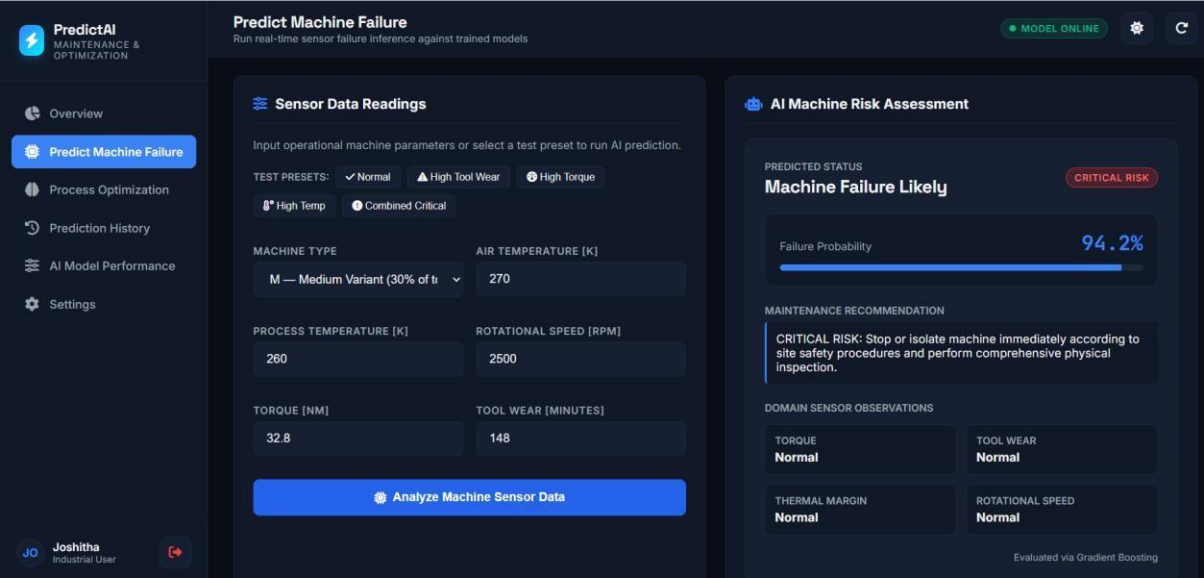
## 6. Snapshots / Screenshots

By accessing the provided application link, the user is directed to the **web application**, where they can register or sign in <https://ai-industrial-maintenance-optimizat.vercel.app/>



The screenshot displays the 'PredictAI' web application interface. On the left, a dark blue sidebar contains the app's logo (a gear icon) and the text 'PredictAI Predict Failures. Prevent Downtime. Optimize Operations.' Below this, a description states: 'Next-generation industrial maintenance and process optimization intelligence for equipment monitoring, powered by Machine Learning & Deep Neural Networks.' A list of features follows: 'Real-time AI failure risk probability evaluation', 'ML vs Deep Learning model performance comparison', and 'Automated maintenance recommendations & process analytics'. At the bottom of the sidebar, it reads 'AI Based Industrial Maintenance & Process Optimization Platform'.

On the right, the registration form is visible. It has two tabs: 'Sign In' and 'Register Account', with 'Register Account' being the active tab. The form includes three input fields: 'USERNAME' (containing 'Joshitha'), 'EMAIL ADDRESS' (containing 'vjoshitha3486@gmail.com'), and 'PASSWORD' (containing masked characters). A blue 'Create Account' button with a user icon is positioned below the password field.



**PredictAI**  
MAINTENANCE & OPTIMIZATION

**Prediction History**  
Search and audit past equipment evaluations

**Sensor Prediction Logs**

Search by model, status, recommendation...

All Risk Levels All Types (L, M, H)

| TIMESTAMP             | VARIANT | AIR TEMP | PROG TEMP | RPM      | TORQUE   | TOOL WEAR | FAILURE RISK | RISK LEVEL | RECOMMENDATION                                                                                                                                                                                                                                                                        |
|-----------------------|---------|----------|-----------|----------|----------|-----------|--------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8/19/2026, 7:52:46 AM | M       | 278 K    | 268 K     | 2580 RPM | 32.8 Nm  | 148 min   | 94.19%       | CRITICAL   | CRITICAL RISK: Stop or isolate machine immediately according to site safety procedures and perform comprehensive physical inspection.                                                                                                                                                 |
| 8/19/2026, 7:52:37 AM | M       | 278 K    | 268 K     | 2580 RPM | 52.8 Nm  | 148 min   | 100%         | CRITICAL   | CRITICAL RISK: Stop or isolate machine immediately according to site safety procedures and perform comprehensive physical inspection.                                                                                                                                                 |
| 8/19/2026, 7:52:25 AM | M       | 278 K    | 268 K     | 2580 RPM | 72.8 Nm  | 148 min   | 100%         | CRITICAL   | CRITICAL RISK: Stop or isolate machine immediately according to site safety procedures and perform comprehensive physical inspection. Inspect drive motor bearings, gearboxes, and mechanical resistance.                                                                             |
| 8/19/2026, 7:51:44 AM | M       | 343 K    | 358 K     | 2580 RPM | 72.8 Nm  | 148 min   | 100%         | CRITICAL   | CRITICAL RISK: Stop or isolate machine immediately according to site safety procedures and perform comprehensive physical inspection. Inspect drive motor bearings, gearboxes, and mechanical resistance.                                                                             |
| 8/19/2026, 7:51:33 AM | M       | 343 K    | 358 K     | 2580 RPM | 142.8 Nm | 148 min   | 100%         | CRITICAL   | CRITICAL RISK: Stop or isolate machine immediately according to site safety procedures and perform comprehensive physical inspection. Inspect drive motor bearings, gearboxes, and mechanical resistance.                                                                             |
| 8/19/2026, 7:51:24 AM | M       | 343 K    | 358 K     | 3580 RPM | 142.8 Nm | 148 min   | 100%         | CRITICAL   | CRITICAL RISK: Stop or isolate machine immediately according to site safety procedures and perform comprehensive physical inspection. Inspect drive motor bearings, gearboxes, and mechanical resistance. Inspect speed control drive, inverter calibration, and mechanical coupling. |
| 8/19/2026, 7:50:29 AM | M       | 298.1 K  | 308.6 K   | 1580 RPM | 42.8 Nm  | 188 min   | 8.62%        | LOW        | Machine parameters healthy. Continue normal operation and routine monitoring.                                                                                                                                                                                                         |

Joshihitha Industrial User

## 7. Challenges Faced

During the development of the Industrial Maintenance and Process Optimization system, several technical challenges were encountered:

- Class Imbalance**  
 Machine failure cases were much fewer than normal cases, making it difficult to detect failures accurately. Therefore, metrics such as Recall, F1-score, and ROC-AUC were considered along with accuracy.
- Model Performance Comparison**  
 Different models produced different results for precision, recall, and ROC-AUC. Multiple models had to be trained and compared to select a suitable production model.
- Categorical Feature Encoding**  
 The machine **Type (L, M, H)** required proper encoding. One-Hot Encoding was selected to avoid introducing an incorrect ordinal relationship.
- Model Integration with Backend**  
 Integrating the trained ML pipeline with the Flask API required ensuring that the input features and preprocessing steps matched those used during training.
- Risk and Recommendation Logic**  
 Converting model probabilities into meaningful risk levels and generating suitable maintenance recommendations required additional rule-based logic.
- Library and Environment Compatibility**  
 Managing Python libraries and their versions during model development and backend integration caused some dependency and environment-related issues.

## 7. Testing and Validation

The complete system required testing of model predictions, API requests, authentication, database storage, and dashboard functionality.

Despite these challenges, **iterative testing, debugging, and model evaluation** helped us develop a stable predictive maintenance and process intelligence system.

## 8. Learnings & Skills Acquired

The development of the AI Based Industrial Maintenance and Process Optimization system resulted in significant technical, analytical, and professional learning outcomes:

### 1. Practical Understanding of Machine Learning

Gained hands-on experience in data preprocessing, feature preparation, model training, and evaluation for machine failure prediction.

### 2. Predictive Maintenance Concepts

Understood how machine sensor parameters can be used to identify potential failures and support proactive maintenance decisions.

### 3. Data Analysis & EDA

Improved skills in exploratory data analysis, correlation analysis, distribution analysis, and identifying patterns in industrial datasets.

### 4. Model Evaluation & Selection

Learned to compare classification models using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix.

### 5. Backend & ML Integration

Gained practical experience integrating a trained machine learning model with a **Flask REST API** for real-time predictions.

### 6. Database & Authentication

Learned to implement database storage, prediction history, user authentication, JWT-based authorization, and password hashing.

### 7. Debugging, Testing & Documentation

Improved troubleshooting, dependency management, system testing, technical documentation, and presentation skills.

Overall, the project strengthened both our AI/ML knowledge and software engineering skills, providing practical experience in developing an end-to-end predictive maintenance and process intelligence solution.

## 9. Testimonials from team

### Team Member 1:

“This project deepened my understanding of machine learning and predictive maintenance; especially how industrial sensor data can be used to identify potential machine failures.”

**Team Member 2:**

"I gained hands-on experience in data preprocessing, exploratory data analysis, feature engineering, and evaluating different machine learning models for failure prediction."

**Team Member 3:**

"The project improved my understanding of backend development and helped me learn how to integrate a trained machine learning model with a Flask REST API and database."

**Team Member 4:**

"This project strengthened my analytical and problem-solving skills through model comparison, process intelligence, risk analysis, and visualization of machine prediction results."

**Team Member 5:**

"Working on this project helped me understand the complete development pipeline, from dataset preparation and model development to deployment, testing, and documentation."

## 10. Conclusion

The AI Based Industrial Maintenance and Process Optimization project demonstrates how artificial intelligence and machine learning can be applied to industrial machine data to support proactive maintenance. The system analyses machine operating parameters, predicts the probability of machine failure, classifies the level of risk, and provides relevant maintenance recommendations.

Throughout the development, the team gained practical experience across the complete pipeline, including data preprocessing, exploratory data analysis, machine learning model development, model evaluation, backend integration, database management, and dashboard analytics. The project demonstrated how raw machine sensor data can be transformed into meaningful predictions and actionable process insights.

Overall, the project provides a practical and industry-oriented solution for predictive maintenance and process intelligence, with potential for future enhancements such as real-time IoT sensor integration, automated alerts, advanced time-series prediction, and improved maintenance scheduling.

## 11. Acknowledgements

We express our sincere gratitude to **Infosys Springboard** for providing us with an excellent learning environment, industry-oriented training, and the opportunity to work on the **AI Based Industrial Maintenance and Process Optimization** project.

We extend our heartfelt thanks to **our mentor** for providing continuous guidance, constructive feedback, and encouragement throughout the project. The guidance helped us understand the practical application of machine learning, improve our approach, and successfully integrate the different components of the system.

We also sincerely appreciate the support of our **teammates, peers** for their motivation, collaboration, and valuable suggestions, which helped us complete the project successfully.